TRUSTIA AI — INVESTOR PRESENTATION

Target: Seed Round ($500,000)

Sovereign Level-4 Autonomous Driving & GPS-Denied Defense UGV Software Stack

Executive Summary: Trustia AI is a sovereign, hardware-agnostic Level-4 autonomous driving intelligence engineered with 16,000+ lines of deterministic C++/Python and verified through 1,301 automated unit/integration tests (%100 pass rate). Our plug-and-play retrofit brain integrates into commercial electric vehicle platforms (Hyundai Ioniq 5 / E-GMP) and tactical military UGVs within 45 days, delivering zero-driver Robotaxi mobility in metropolises and sub-3cm 3D LiDAR SLAM navigation in GPS-denied combat zones.

Figure 1: Trustia AI Hyundai Ioniq 5 Level-4 Test Platform

Parameter	Accreditation & Data
Founder & Architect	Murat Furkan Bayram (17)
Defence Industries Bsk.	SSB 100/100 Perfect Score (L2zPtN4X1ZJ)
Incubation Base	Istanbul Chamber of Commerce (ITO) BTM
Govt. R&D; Certification	KOSGEB Advanced Entrepreneur & TUBITAK
Automated Testing	1,301 / 1,301 Verified Tests (100% Pass)
Target Vehicle	Hyundai Ioniq 5 CAN-FD & Tactical UGVs

Institutional Certifications & Metrics

1. Level-4 Robotaxi Reduces driver costs to 0% and elevates fleet net operating margins to 72%.	2. GPS-Denied SLAM Navigates in electronic warfare environments with millimeter 3D LiDAR mapping.	3. 45-Day Turnaround Retrofits stock EVs into driverless robotaxis without building custom chassis.

INDUSTRY BOTTLENECKS

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MARKET PAIN & SOLUTION ARCHITECTURE

Target: Seed Round ($500,000)

Current Market Pain & Bottleneck	Trustia AI Breakthrough Solution
High Driver Labor Costs: Human driver wages and insurance account for 65-75% of ride-hailing operational expenses in major metropolises.	0% Driver Labor & 72% Margin: Level-4 autonomy allows 24/7 continuous operations with high gross profit margins.
GPS-Denial Vulnerability: Electronic jamming in tactical zones or tunnels blinds standard autonomous vehicles, causing mission failure.	GPS-Denied 3D LiDAR SLAM: Multi-layer LiDAR Odometry & Error-State Kalman Filter (ESKF) provides sub-3cm localization without GPS.
Foreign Black-Box Software Dependency: Proprietary foreign stacks create national security vulnerabilities and data leak risks.	100% Sovereign Deterministic Code: 16,000+ lines of native, verifiable C++/Python code free from external black-box dependencies.
Long 5-Year OEM Development Cycles: Building bespoke autonomous vehicles from scratch requires $100M+ and 4-5 years.	45-Day Plug-and-Play Retrofit: Standard CAN-FD and ROS2 bridges transform existing production EVs into autonomous robotaxis in 45 days.

Trustia AI 4-Tier Technical Architecture

1. Perception & Fusion (100 Hz)	Microsecond temporal synchronization of Ouster 128ch LiDAR, Livox Mid-360, Continental 77GHz Radar, and Sony GMSL2 HDR cameras.
2. 3D SLAM & State Estimation	Real-time pointcloud registration (NDT/ICP) tightly coupled with 400 Hz IMU for drift-free metric mapping in GPS-denied environments.
3. Trajectory & Behavior Planning	Hybrid A* kinematic path planner, dynamic B-Spline generation, 5-second VRU intention forecasting, and roundabout intersection negotiation.
4. Deterministic Controller (RTOS)	Pure Pursuit / Model Predictive Control injecting steering torque and braking commands via CAN-FD at 100 Hz with ASIL-D safety watchdog.

TEST VEHICLE GALLERY

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HYUNDAI IONIQ 5 LEVEL-4 HARDWARE INTEGRATION

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Physical Test Vehicle & Cockpit Integration Gallery

Figure 2: Side Profile (360 Roof LiDAR & Lateral GMSL2 Cameras)

Figure 3: Cockpit Autonomous C2 Touchscreen & E-Stop Console

Figure 4: Front Grille (Livox Bumper LiDAR & Radar Integration)

Figure 5: Rear Profile (Dual GNSS Antennas & 5G Teleoperation)

Hardware Integration Note: The sensor suite mounts securely to OEM roof rails and bumper air intakes without drilling body panels. An isolated sub-trunk electrical harness houses a dedicated LiFePO4 power rail, 80A automotive relay, and physical Schneider Electric E-Stop.

HARDWARE BILL OF MATERIALS

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LEVEL-4 SENSOR & HARDWARE SPECIFICATIONS (BOM)

Target: Seed Round ($500,000)

Hardware Subsystem	Specification & Model	Functional Role	Interface Protocol
Primary Roof LiDAR	Ouster OS2-128 Rev7 (128 Channels, 200m)	360 Long-range 3D pointcloud dense mapping	Gigabit Ethernet / UDP
Perimeter LiDARs	2x Livox Mid-360 (Front & Rear Bumpers)	Close-range blind spot, curb, and hazard detection	100BASE-TX Ethernet
Automotive Radars	2x Continental ARS 408-21 (77 GHz)	All-weather relative velocity & range tracking	CAN-FD / DB9
Vision Perception	4x Sony IMX390 HDR (GMSL2, 120dB)	Traffic light, lane boundary & pedestrian recognition	GMSL2 / FAKRA-Z
AI Compute Brain	NVIDIA Jetson AGX Orin 64GB (275 TOPS)	Real-time sensor fusion, SLAM & MPC control	PCIe Gen4 / RTOS Linux
Drive-by-Wire Bridge	Kvaser U100 CAN-FD DB9 Isolated Interface	Direct E-GMP LKAS/SCC torque command injection	CAN-FD (ISO 11898-2)
Connectivity & C2	Teltonika RUTX50 Industrial 5G Router	Cloud teleoperation & real-time telemetry link	5G / Dual-SIM / IPsec

ISO 26262 ASIL-D Functional Safety Architecture: Features a 5ms immediate human driver takeover override, a 200ms autonomy watchdog, and an automated Minimum Risk Maneuver (MRM) that brings the vehicle to a safe stop in emergency pull-over lanes upon subsystem degradation.

REVENUE STREAMS & SCALING

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BUSINESS MODEL & UNIT ECONOMICS

Target: Seed Round ($500,000)

1. Robotaxi Mobility-as-a-Service (MaaS)	2. OEM & Fleet Software Licensing (SaaS)
• Model: Direct passenger transport in geofenced corridors (Maslak-Levent-Airport). • Daily Volume: ~18 rides/day at $10.50 average ticket. • Monthly Gross: ~$5,500 gross revenue per vehicle. • Net Margin: 72% Net Operating Margin due to 0% driver labor wages.	• Model: Software brain licensing to taxi fleets, logistics, and rental operators. • Integration Fee: $10,000 per vehicle one-time installation license. • Recurring SaaS: $350/month per vehicle for live HD map & software updates. • Scaling: 50 licensed vehicles deliver $650,000 pure annual profit.
3. Defense & Tactical UGV Retrofits (B2G)	**4. Scalable High-Margin Unit Economics**
• Model: Sovereign GPS-denied autonomy retrofits for defense prime contractors. • Contract Value: $150,000 – $450,000 per project PoC and software licensing. • Retention: Annual STANAG 4586 tactical software maintenance contracts.	• 1 Vehicle (Pilot): ~$65,000 net annual cash flow. • 10 Vehicles (Micro Fleet): ~$650,000 annual net revenue. • 100 Vehicles (City Scale): ~$6.5 Million annual gross revenue. • Asset-Light: Scales through fleet partner vehicles without heavy CapEx.

FINANCIAL PLAN & P&L;

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3-YEAR FINANCIAL PROJECTIONS & MILESTONES

Target: Seed Round ($500,000)

Financial Metric (USD / Units)	Year 1 (2026-27)	Year 2 (2027-28)	Year 3 (2028-29)
Active Robotaxi Fleet	2 Vehicles (Pilot)	15 Vehicles	80 Vehicles
Licensed 3rd-Party Vehicles	5 Vehicles	40 Vehicles	250 Vehicles
Defense PoC Contracts	1 Contract	3 Contracts	6 Contracts
Total Gross Revenue ($)	**$250,000**	**$1,250,000**	**$5,500,000**
Gross Profit Margin (%)	68%	74%	81%
Operating Expenses (OPEX)	$125,000	$420,000	$1,100,000
EBITDA ($)	**$62,000**	**$550,000**	**$2,750,000**
Engineering Headcount	6 Engineers	14 Engineers	28 Engineers

Growth Milestones & Expansion

Months 0 - 6: Complete Hyundai Ioniq 5 Level-4 kit, closed-track validation at BTM Fulya & Bilisim Vadisi, first VIP live demos.	Months 6 - 12: Launch Maslak-Fulya commercial pilot corridor, deliver defense supplier PoC, achieve initial recurring SaaS cash flow.
Months 12 - 24: Scale fleet to 15 vehicles via taxi partnerships, secure global scaling rounds (EIC Accelerator & Y Combinator).	Months 24 - 36: Expand to 80+ Robotaxis in Turkey and license autonomy stack across Eastern Europe & GCC region ($5.5M+ ARR).

SEED ROUND TERMS

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INVESTMENT ASK & USE OF FUNDS

Target: Seed Round ($500,000)

Funding Target: $500,000 Seed Round (10% Equity at $5,000,000 Post-Money Valuation)

Allocation Category	Share	Amount ($)	Key Deliverables & Milestones
Vehicle & Sensor Hardware	35%	$175,000	2x Hyundai Ioniq 5 test vehicles, Ouster 128ch LiDARs, Orin kits.
Senior Engineering Talent	40%	$200,000	5 Senior robotics, embedded systems & SLAM engineers (24-month runway).
Field Testing & Sandbox	15%	$75,000	BTM / Municipality pilot permits, closed track testing, safety audits.
Working Capital & Legal	10%	$50,000	International PCT patent filings, cloud compute, and operations.
TOTAL CAPITAL	**100%**	**$500,000**	**24 Months Runway to Breakeven & Commercial Scaling (Month 14)**

Target Term Sheet Summary:
- Investment Amount: $500,000 (Five Hundred Thousand USD)
- Valuation: $5,000,000 Post-Money Valuation for 10% Equity.
- Investor Rights: Lead investor receives Board Observer seat and quarterly information rights.
- Leverage: Private capital is matched with non-dilutive state grants (KOSGEB + TUBITAK ~$120k).

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